FYP-2 Mid Evaluation — Presentation Plan and Speaking Scripts

Project S26-017-R-GASR — 4D Surgical Scene Reconstruction and Deformation Prediction

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Deck: **FYP2_Mid_Eval_Modules_5_and_6.pptx** — 23 slides, of which **7 are presented**

Built to the FYP Committee's mandated structure for a research-based FYP

1. The Committee's Rules, and How This Deck Answers Them

The committee allows **4 to 5 minutes of uninterrupted presentation**, and prescribes exactly how those minutes are spent. The rest of the slot is live demonstration. The deck is built in three parts so that the first seven slides *are* the mandated five minutes, with nothing extra to skip past.

Required section	Time	Slides	What covers it
Overview of FYP goals and objectives	1 min	1 – 2	The goal in one sentence, four objectives each with an explicit status, and the research question stated as a question.
Address faculty feedback from the FYP-1 final evaluation	1 min	3	Four feedback points, each paired with what was built and the file that proves it.
Progress since the FYP-1 final evaluation	2 min	4 – 6	Methodology finalised, metrics and results, and where we stand against the literature.
Remaining work until the final FYP-2 evaluation	1 min	7	Five items in dependency order, each marked implemented or outstanding.
Demonstration + preliminary results	rest	8 – 12	Live inference in Colab, then the result slides to describe what it produced.
— held for questions —	—	13 – 23	Architecture, both modules in detail, problems and fixes, training curve, literature table.

The five R&D failure modes, and where each is answered

The committee listed five reasons research FYPs fail. Each one has a slide pointed at it. If a panel member is working from that list, you want to be able to answer by turning to a slide.

Stated failure mode	Where this deck answers it
Not finalising and reporting the methodology	**Slide 4** states the problem formulation, the model, the target, the sampler and the data protocol, and names the manuscript section where it is written up in full.
Not choosing valid metrics of evaluation	**Slide 5, left panel** gives five metrics and *why each was chosen* — including why Chamfer Distance alone is insufficient.
Not reporting adequate results	**Slide 5, right panel** and **slides 10 – 12**: accuracy, direction, uncertainty, the full baseline table, and the two explanations we tested and eliminated.
Not comparing results with current literature	**Slide 6** compares against three published reconstruction systems and two prediction base papers; **slide 20** holds the full four-field gap table.
Not having an impressive presentation	Seven slides for five minutes, a live demo rather than a video, and a reference section so no question ends in "we would have to check".

> **Three rules from the committee that cost marks if missed.**
>
> - **One laptop.** Put the deck, the notebook, the checkpoint and the rendered outputs on a single machine before the day. No switching.
> - **Demo videos are not accepted.** The demonstration must be the live system. Slide 8 says out loud that nothing is pre-rendered — only say that if it is true.
> - **The 4–5 minutes are uninterrupted.** Do not stop for questions during slides 1 to 7, and do not overrun — every minute you take from the demo is a minute of marks.

2. Who Presents What

Each member presents one unbroken run. The mandated five minutes divides along the committee's own time allocation, which happens to match the subsystem each of you owns.

Member	Slides	Time	Their part
Ahmad Rashad 22i-1175	1 – 3	≈ 2 min	Opens. Goals and objectives, then the FYP-1 feedback and what was done about it. Also answers architecture and model questions from the reference section.
Shehroz 22i-0771	4 – 6	≈ 2 min	The whole progress section: methodology, metrics and results, and the literature comparison. Every number in it came out of his evaluation framework.
Humair Khan 22i-2623	7 – 12	≈ 1 min + demo	Remaining work, then drives the live demonstration and narrates slides 8 to 12 as it runs. The rendering and visualisation being demonstrated are his.

Two handovers in the mandated block, after slide 3 and after slide 6. The person handing over steps back; the next steps in. Do not pass the clicker mid-sentence.

3. Ahmad Rashad (22i-1175) — Slides 1 to 3 · 2 minutes

Slide 1 — Title · 10 seconds, no more

Say this while the deck is already up. Do not spend presentation time on a title slide.

Good morning. I am Ahmad Rashad, with Shehroz and Humair. Our supervisor is Dr Akhtar Jamil, and this is the FYP-2 mid evaluation for project S26-017, a research project on 4D surgical scene reconstruction and deformation prediction.

Slide 2 — What the project is for · 1 minute

On screen: goal statement, four objective cards with status, the research question. This is the committee's "overview of goals and objectives" minute. Stay on goals — do not drift into method.

The goal in one sentence. A surgeon operating through a single laparoscope sees a flat image of the present moment. We want a system that reconstructs that scene in 3D and then anticipates how the tissue will deform next — from that one camera, with no depth sensor and no clinical annotation.

Walk the four objective cards left to right. Say the status of each out loud — the panel is marking progress, so the statuses are the point of the slide.

Four objectives. Reconstruct the scene in 3D from monocular video — complete, that was FYP-1. Learn a self-supervised model of how tissue deforms — complete, that is Module 5. Predict future geometry at one to three frames with a calibrated confidence estimate — implemented, that is Module 6. And evaluate it against classical and learned baselines and against the literature — in progress, Modules 7 and 8.

Pause, then read the research question slowly. It is the sentence the panel should remember.

Which gives us the research question: can future 3D tissue geometry be predicted from monocular surgical video alone — and can the system say how much to trust each prediction?

This is a research FYP, so what we are delivering is a methodology, a set of measured results and a comparison against the literature.

Slide 3 — Faculty feedback from FYP-1, and what we did about it · 1 minute

On screen: four feedback → action rows, each naming the source file. This is a required section and it is the easiest minute in the talk to score on, because every row has evidence. Do not editorialise — say the feedback, say what was built, name the file.

We were given four pieces of feedback after FYP-1, and all four are implemented.

First, implement a base paper for comparative analysis. We implemented two: PointRNN by Fan and Yang, which is the closest published method to ours — future point clouds from past ones, scored with Chamfer Distance — and Luo and Hu's CVPR 2021 paper, the first diffusion model over 3D point clouds. Both are in module7_baselines.

Second, select a few state-of-the-art papers and implement them. We built eleven methods on a single protocol: five classical predictors, five learned ones, and ours, all scored on identical windows. That is 781 lines in the same file.

Third, do a detailed analysis — hyperparameter tuning and optimisation. Module 8 runs four studies: inference cost against accuracy, context length, optimiser sensitivity, and diffusion ablations.

Fourth, start writing a paper in IEEE or Springer format. There is a six-page IEEE conference manuscript in the repository with 22 references, and the tables in it are generated directly from the result files rather than typed in by hand.

Say the honest caveat. It costs you nothing here and it buys credibility for the rest.

All four are written and in the repository. Two of them — the eleven-method comparison and the tuning studies — still need their compute runs, and that is the first item on our remaining-work slide.

Handover. Step back from the laptop.

Shehroz will take you through the progress since FYP-1.

> **Ahmad — if asked during questions**

- **Architecture, the network, why diffusion, training, overfitting, hyperparameters** — slides 14 to 16 and 19. Jump to the architecture slide and trace the two bands.
- **"Why is the model only 16.3 M parameters?"** — The corpus is ten short videos. A larger model overfits faster; we measured overfitting from epoch 60 as it is.
- **"Why predict velocity rather than noise?"** — At a displacement scale of 0.004, noise prediction is numerically badly conditioned. The v-parameterisation stays well conditioned across the schedule.
- **"Did you address all of Dr Jamil's feedback?"** — Implemented yes, fully evaluated no, and slide 3 says which two are waiting on compute.

4. Shehroz (22i-0771) — Slides 4 to 6 · 2 minutes

The committee's "progress since the FYP-1 final evaluation" section. Three slides, forty seconds each. This is the densest part of the talk — rehearse it against a timer.

Shehroz — the tone to hold. Three of the committee's five research failure modes are decided in your two minutes: methodology, metrics, results. Say what you built, what you measured, and what it came out as — including the part where we do not beat the baseline. A panel marks an R&D project on whether the measurement is sound, not on whether the number is flattering.

Slide 4 — The methodology, finalised · 40 seconds

On screen: problem statement in the mint bar, three teal cards, two gold cards. Do not read the cards out — say the four things below and let the slide carry the detail.

Thank you, Ahmad. The methodology is now fixed, and this is it in one line: given the last four frames of 512 tracked tissue points, predict their displacement k frames ahead, for k of one, two or three — with no clinical labels at any stage.

The model is a conditional denoising diffusion model, 16.3 million parameters, using v-prediction on a cosine schedule with Min-SNR weighting, and a temporal transformer with two-way cross-attention.

The middle card is the one design decision worth naming aloud.

One decision matters more than the rest. It does not predict the displacement directly — it predicts the *correction* to constant-velocity extrapolation. That means a model that learns nothing falls back to an estimator that is already 41% better than the naive baseline, rather than falling back to the naive baseline itself.

We query it three ways: directly for one to three frames, by rollout to ten frames, and by drawing five independent samples whose spread becomes the uncertainty estimate.

Ten videos, 5,910 training and 1,461 validation windows, self-supervised throughout — the targets come

from the reconstruction, never from an annotator. The full method is written up in Section III of the IEEE manuscript, with every hyperparameter stated.

Slide 5 — Metrics chosen, and what they measured · 45 seconds

On screen: metrics and the reason for each on the left; three result cards on the right; the eliminated explanations along the bottom. Lead with why the metrics, because that is what is being marked.

Metrics first, because the choice matters. Chamfer Distance is our primary accuracy metric because it is correspondence-free — it does not assume point identity survives across frames. We report Mean Point Error alongside it for the correspondence-aware view.

This next sentence is the methodological point of the slide. Do not rush it.

But Chamfer Distance alone cannot separate a model that learned nothing from a model that learned the right direction and the wrong distance. Both score identically. So we added a directional cosine, which separates them, and a spread–error correlation to test whether the uncertainty is calibrated.

Now the three result cards, right side.

And here is what they measured. On accuracy we are not yet ahead: Chamfer Distance 0.01568 against a naive baseline of 0.01557, and constant velocity beats both of us, on all ten videos.

On direction, the model has clearly learned something: cosine plus 0.57 at one frame ahead, plus 0.40 at two, plus 0.26 at three. A model that learned nothing would read zero. So our failure is magnitude, not direction — and that is a far more tractable problem.

And on uncertainty, the spread across five sampled futures correlates with realised error at 0.69. No other method in our comparison provides that at all.

Bottom bar — ten seconds, but it is the line that shows the work was done properly.

Before concluding any of that we tested and eliminated two explanations: error does not grow with distance from the training frames, and dropped preprocessing frames do not explain it — 333 origins checked, zero drops.

Slide 6 — How we compare with published work · 40 seconds

On screen: reconstruction PSNR table on the left, the two prediction base papers on the right. This is the committee's "comparison with current literature" requirement.

Against the literature, on both halves of the project.

For reconstruction, the base paper family is LerPlane, EndoGaussian and Deform3DGS, which report 29.46, 30.25 and 30.48 dB. We report 30.10.

State the caveat yourself. Do not let the panel find it.

That is indicative, not controlled, and we say so in the paper: those figures use stereo supervision on different sequences. What it does establish is that we are within 0.4 dB of the stereo state of the art using

a single camera and no depth sensor — and that geometry is what Module 5 learns to predict.

For prediction, our two base papers are PointRNN, implemented as our primary learned baseline, and Luo and Hu's point-cloud diffusion, of which our denoiser is a conditional, temporally grounded instance.

The full comparison is eleven methods on one protocol. It is implemented, and the run is the next step.

Handover. Step back.

Humair will cover what is left, and then show you the system running.

> **Shehroz — if asked during questions**
>
> - **"So your model does not work?"** — It does not beat the baseline on Chamfer Distance and we report that. It learns direction at 0.57 and produces calibrated uncertainty at 0.69. We have localised the cause to magnitude and implemented two corrections for it.
> - **"Why not just use constant velocity?"** — For raw accuracy at one frame, today, you should. It gives you no uncertainty, no multiple futures and no route to improvement.
> - **"How do you know the evaluation itself is sound?"** — QT0. The rebuilt trajectory is verified against Module 4's saved windows and runs as a gate before anything is predicted. Slide 18 has it.
> - **"Did you validate your baselines?"** — Yes, and two were wrong when we checked. Constant acceleration was taking velocity at t − ½; the "Kalman filter" was an alpha–beta filter. Both fixed and verified against closed-form cases. Slide 18.
> - **"Is 0.69 good?"** — For an uncertainty estimate never trained against an error signal, yes. It is emergent from sampling, not fitted.

5. Humair Khan (22i-2623) — Slide 7, then the demonstration

One mandated minute on remaining work, then you drive the live system for the rest of the slot. Slides 8 to 12 are your narration while it runs.

Slide 7 — Remaining work until the final evaluation · 1 minute

On screen: five numbered rows in dependency order. The committee asks for this explicitly. Give each item a status, not a hope.

Thank you, Shehroz. Five things remain, in the order they have to happen.

One: run the eleven-method comparison and the tuning studies. Both modules are written and tested — what is left is compute, not code.

Two: evaluate the two corrections we have already implemented. Magnitude calibration needs no retraining at all; the residual target needs one retrain.

Three: move to leave-one-video-out evaluation, so we are not selecting and reporting on the same recordings.

Four: fold the completed results into the manuscript. The draft is written and its tables are generated from result files, so finished runs populate it directly.

Five: submit, and extend the corpus beyond ten recordings — the overfitting we measured is consistent with a dataset this size.

Move straight into the demo. Do not pause for questions — the five minutes are uninterrupted.

And with that, let me show you the system running.

Slide 8 — What you are about to see, running live · while the notebook loads

Put this up and start the notebook behind it. Talk over the loading — never watch a progress bar in silence.

Everything from here is live. Nothing is pre-rendered: the checkpoint is loaded and inference runs in front of you.

Five steps. We load the trained checkpoint and verify the evaluation data — QT0 runs first and checks the rebuilt trajectory against Module 4's saved windows before anything is predicted. We predict a held-out moment at one, two and three frames ahead, against the naive and constant-velocity baselines on the same window. We render that prediction as an actual video frame, which means carrying the predicted motion of 512 points onto all 58,000 Gaussians. We compare it side by side with the real frame. And then we forecast past the end of the recording.

Slide 9 — Making the prediction visible · narrate while rendering runs

Your own subsystem. Use this slide while the render cell is executing.

The difficulty here is a mismatch. A prediction moves 512 tracked points; a picture needs all 58,000 Gaussians. The bridge is a distance-weighted interpolation, weighted so a tracked point receives its own predicted motion exactly rather than an approximation of it.

Rendering it meant re-implementing Module 2's renderer inside the prediction module — positional encoding, deformation network and degree-3 spherical harmonic colour. I verified that against the original rather than assuming the two matched.

Two honesty points. Make them before anyone asks — they are what distinguishes a demonstration from a demo reel.

Two things about what you are seeing. Predicted motion at one to two pixels is invisible to the eye, so motion is drawn as arrows coloured by agreement with the truth and magnified automatically — and the magnification factor is printed on every panel and never applied to any reported number.

And second, until recently nothing on screen was actually a prediction — every predicted frame was paired with a frame that already existed. The forecast now runs twelve frames past the end of the recording. The real-video panel stops and says no footage exists beyond that point; the predicted panel keeps going. That is the only output that shows prediction rather than describing it.

Point at the three PSNR tiles once the render appears.

In image space: the reconstruction ceiling is 30.04 dB — the best any prediction could look. Ours is 26.79. Naive is 26.74.

Slides 10 to 12 — describing the results · as the demo produces them

These three are the committee's "preliminary results properly visualised and described". Shehroz can take them if a question turns technical — agree which of you narrates before you start.

Slide 10, the key finding: the model knows which way tissue moves, not how far. Cosine plus 0.57 at one frame ahead, decaying to zero by five. A model that learned nothing would read zero everywhere.

Slide 11, uncertainty: when the five sampled futures disagree with each other, the prediction really is worse — correlation 0.69. A system that is confidently wrong is unusable in theatre; one that can say "do not trust me here" is usable even when imperfect.

Slide 12, the full comparison: our model against the naive baseline and constant velocity at all three horizons, with rollout stability and the two eliminated explanations.

> **Humair — demo checklist, to be done before the day, not on it**
>
> - Notebook open, runtime already connected, GPU confirmed. A cold Colab start in front of a panel is a lost minute.
> - Checkpoint and workspace already mounted from Drive and **verified loadable** — run the whole demo path end to end at least twice the day before.
> - Every heavy cell run once already so the outputs are cached, and the cells you will run live timed. Know which cell takes longest and what you will say over it.
> - A fallback: if the runtime dies, have the already-executed notebook with its outputs visible. That is still the live system, just not re-executed — say so plainly rather than pretending.
> - Deck, notebook, checkpoint and rendered outputs all on **one laptop**, as the committee requires.
> - Screen resolution set so the notebook text is readable from the back of the room.

6. Question Routing

Agree this before you walk in, so three people do not answer at once. Whoever is asked something that belongs to someone else passes it in one sentence.

If the question is about…	It goes to
Goals, scope, the network, the diffusion formulation, training, overfitting, hyperparameters, the architecture, how feedback was addressed	**Ahmad** · slides 14–16, 19
Methodology, metrics, baselines, the quality tests, evaluation integrity, the	**Shehroz** · slides 17–18, 20

comparison, the cosine finding, uncertainty, statistics

The demo itself, rendering, the forecast, reconstruction quality, the dataset, the runtime environment — **Humair** · slides 9, 21

Timeline, division of work, what remains — **Humair** · slide 7

The one question to rehearse as a team. "Your model does not beat a two-line baseline — why should this pass?" Agree on one answer and let one person give it. Suggested: *"Because we can tell you exactly why it does not, and the diagnosis is itself a result. We measured that the model learns motion direction at 0.57 and fails only on magnitude; we eliminated drift and dropped frames as explanations; and we implemented the two corrections that follow from that. We also produced a calibrated uncertainty estimate that none of the baselines can give you. For a research project we would rather bring you a measured negative with an identified cause than an unexplained positive."*

Timing rehearsal. Run the mandated block against a stopwatch at least three times. The target is 4 minutes 50 seconds for slides 1 to 7. If you are over, the two places to cut are the objective statuses on slide 2 (name only the two that changed since FYP-1) and the metric rationale on slide 5 (keep Chamfer Distance and the cosine; drop the other three). Do not cut slide 3 — it is a required section.